

Software Engineering, Spring 2026

PROJECT DESCRIPTION

Project Portfolio Web Platform

Milestone 1: Grade 15% - Deadline 14 March, 2026 11:59pm

Milestone 2: Grade 15% - Deadline 11 May, 2026 11:59pm

Assignment: Grade 10% - TBD

Please read the following instructions carefully:

- Any case of **plagiarism or cheating** in Milestone 1, Milestone 2 or the assignment, will result in a zero.
- It is **YOUR responsibility** to ensure that you have:
 - Read and understood **everything** in the project description (this document).
 - Submitted before the deadline.
 - Submitted the correct file for Milestone 1 and the assignment on your GitHub repository.
 - Made commits for Milestone 2 on your GitHub repository.

1 Theme

You are required to build a Project Portfolio Web Platform for GUC students that can act as a central platform to showcase the good work that all GUC students finish in their lab courses and bachelor projects.

Note that no code is uploaded directly to the platform itself.

The project includes 2 milestones and an assignment:

- Milestone 1: requirements engineering
- Milestone 2: front-end design and development using React
- Assignment: usage of design patterns in the project

2 Case Description

Refer to the following case description to understand how the web application should work:

1. Ahmed just finished building a web app with his team for his Software Engineering course. Normally, he'd create a shared Google Drive folder or a GitHub organization to keep everyone in the loop — but managing access, and contributions across different tools is messy. He wants a single place where he can set up the project, bring his teammates on board as collaborators, and clearly label it as a course project or bachelor project so it's properly organized from day one, all without uploading the code to the .
2. While working on the project and after wrapping up their first sprint, Ahmed's teammate Sara wants to document what they built. She'd usually update the GitHub README, maybe deploy a live demo on Heroku, and share a screen recording on Google Drive — all scattered across different platforms with no coherent presentation. Instead, she wants one place where she can write a clear project description, link the project web app or website, link their GitHub repo, and attach a demo video so anyone reviewing the project gets the full picture. For their bachelor project, she also needs to upload their thesis without having to email it separately to their supervisor.
3. Once the team is happy with the project, one of the collaborators wants to make it visible to the outside world. Right now their only option would be to post it on LinkedIn and hope the right people see it, or manually share links with potential employers. They want to simply flip a switch on the platform to make the project public, putting it in front of employers without any extra effort.
4. Youssef, a second-year student, wants to see what senior students built in courses he's about to take. He'd normally dig through GitHub profiles, ask around, or

hope someone posted on LinkedIn — with no guarantee the work is relevant or academic. He wants to browse through a curated space of real student projects, filtered by course or project type, so he can get inspired and set a benchmark for his own work.

5. Youssef stumbles across an impressive bachelor project and wants to go deeper. On LinkedIn he'd only see a short post. Here, he can read the full project description, watch the demo, download the thesis, and even read any research papers the student published — giving him a complete academic and technical picture of the work in one place.
6. A tech recruiter at a software company is looking for fresh talent. She currently scrolls through LinkedIn and GitHub profiles, piecing together a student's skills from scattered posts, repos, and CVs that may be months out of date. On the platform, she can browse structured student portfolios that bring together all their best projects in one place, each with demos, descriptions, and GitHub links. When she finds a student whose work stands out, she can reach out directly through the platform without needing to hunt down their contact information elsewhere.

3 Overview

In every single point presented earlier, there are problems that the stakeholders face when using platform. You need to construct a web application to help **students, course instructors, employers and administrators** with their own problems, all while covering **at least** the following modules with their corresponding sub-modules:

1. **Authentication & User Management:**
 - User Registration & Login
 - User Portfolio creation and management
2. **Project Management:**
 - Instructor/collaborator Assignment
 - Feedback and review workflow
 - Notifications
 - Project Creation and Submission
3. **Discovery & Exploration:**
 - Project Search and Filtration
 - Portfolio Search and Filtration

In addition to the case description above, you can also ask yourselves:

- Will all collaborators in a project have the same roles or permissions?

- Will a project (course or bachelor) always be public (for all to view) at all times?
- Who will add the courses students will choose from? And how are the course instructors assigned to these courses?
- What should a portfolio look like and how does it differ from any other project?
- In the case of a bachelor project, how will the supervisors (who is also a course instructor) be added/ assigned to the bachelor project?
- What can employers do on the system?
- How can students be motivated to use the system?
- How can the system make projects more "collaborative", i.e. allowing peers to give feedback on different projects?

3.1 Product Managers

For **both Milestones 1 and 2 and the assignment**, you can ask the **Product Manager** assigned to your team (according to the **Teams** file on the CMS) questions to gather information alongside your own research. Stick to your Product Manager to avoid receiving conflicting responses throughout your project.

3.2 Milestone 1

Milestone 1 will focus on **Requirements Engineering**, where you are required to write a set of complete and consistent **functional and non-functional requirements** for a **web application** based on the modules and sub-modules listed previously in the form of **user stories**. The requirements should be documented, actionable, measurable, testable, traceable, related to identified software needs, and defined to a level of detail sufficient for a developer to understand. Refer to PA1 for help on how to write user stories.

3.3 Milestone 2

Milestone 2 will act as the prototype of your system, which you will be presenting to your Product Manager. Your prototype will be for your web application and will be made up of front-end design using **React**. Note that this means there will be **no database or back-end programming**. To simulate information coming from a database, simply hard code the information (as dummy data) to appear as if it were retrieved.

It is also important to understand that **UI/UX evaluation is purely subjective and based on the opinion of your Product Manager**.

The requirements for Milestone 2 (FE development) will be posted later.

3.4 Assignment

The assignment will require teams to draw a full class diagram of the entire project with the use of design patterns. The details of the assignment will be posted later.

3.5 GitHub Repository

Each team will be given a GitHub repository (repo) to use for this project to submit **Milestone 1, Milestone 2 and the assignment.**

- Each team leader (SCRUM Master) will be granted access to the repo created for their team.
- Team leaders **MUST** accept the repo invitation **BEFORE** the Milestone 1 deadline. Failure to accept the invitation will result in your inability to submit MS1 via GitHub on time, which will result in a **ZERO** in MS1.
- Each team leader **must** grant their other team members access to the repo.

4 Objectives

4.1 Milestone 1

- Learn how to analyze a problem scenario.
- Learn the process of requirements engineering and deriving requirements' specifications.
- Learn to research the whole scope of a given task using any relevant content. For example, textbook(s), Doctors, TAs, and online resources (like [Product Backlog Example](#), etc.).

4.2 Milestone 2

- Learn how to use React to create a simple FE design.
- Collaborate as a team on the same repository on GitHub.
- Follow and implement the UI/UX rules taken in the lectures and tutorials.
- Follow the guidance for the FE design presented by the Product Manager assigned.

4.3 Assignment

- Learn how to draw a full scale class diagram of a full system.
- Learn to identify design patterns that are suitable for the application.
- Learn how to apply design patterns.

5 Deliverables

5.1 Milestone 1

In Milestone 1, you are required to submit a filled Excel sheet of well-written **user stories** that covers the following:

1. All stakeholders of the system (as mentioned in the description).
2. **100 functional requirements** as user stories.
3. **10 non-Functional requirements** of the system. For each one you must highlight if it maps directly to other **functional requirements**. In case it does, then point these functional requirements out. Otherwise, provide criteria on how to measure it.
4. Too many similarities between requirement submissions from different teams will be considered a cheating case and will result in a ZERO for the entire Milestone.
5. **The requirements sheet must follow the specification outlined in the Excel sheet named "RequirementsTemplate.xlsx" and must be submitted on your team's GitHub repository. For Milestone 1, it does not matter who uploads the Excel sheet onto the GitHub repository; we will not be checking commits in Milestone 1.**

ANY SUBMISSIONS OF MS1 BY EMAIL WILL NOT BE ACCEPT. ALL MS1 SUBMISSIONS MUST BE ON THE REPO CREATED.

Also note that in the template, there are examples of functional and non-functional requirements to show you the format of how to write requirements. **These examples are not related to your project.**

5.2 Milestone 2

In Milestone 2, you are required to design and implement **ONLY** the front-end of the platform using React. You may use hard-coded dummy data to represent any data that would otherwise be retrieved from a database.

1. You must make sure your entire project is on the GitHub repositories created for your teams, and each team member **MUST** have commits on the repository.

TEAM MEMBERS WITH NO COMMITS AT ALL WILL RECEIVE A ZERO FOR ALL OF MILESTONE 2.

2. Too many similarities between FE designs will be considered a cheating case and will result in a ZERO for the entirety of Milestone 2.

5.3 Assignment

In the assignment, you are required to draw the class diagram of the entire project, while using the appropriate design patterns. More details will be posted about the assignment later.

6 Grading Criteria

6.1 Milestone 1

The grading criteria of Milestone 1 will depend on the following points and any unfulfilled requirement will result in a grade deduction:

- You must have a **minimum of 10 well-defined** non-functional requirements. These non-functional requirements must be suitable for the system at hand.
- You must have **exactly 100 unique user stories**. If you write more than 100 requirements, we will only look at the **first** 100 requirements in your submission. Any requirements after requirement number 100 **will be ignored**. Therefore, be sure not to have any duplicated requirements.
- For each user story to be considered valid and counted towards your requirements, it has to follow the following rules:
 - The CRUD (create, read, update, and delete) of any user story will be considered as **only one** user story.
 - User stories with the same action and intent but different stakeholders should be written as "As a stakeholder1/stakeholder2/stakeholder3, I should be able to [action] so that I can [rational]". This will be considered **one** requirement of the 100 requirements.

The grading of this assignment will be broken down alongside the previously mentioned specifications as follows:

- All requirements must be as clear and as detailed as possible.

- All requirements must be organized in a proper order with a proper flow.
- All requirements must have dependencies,
- All requirements must be written in correct English.
- All functional requirements must be written as user stories.
- All requirements should **NOT** mention the UI; e.g. click, scroll, homepage.

6.2 Milestone 2

- Milestone 2 is **NOT** dependent on your Milestone 1 submissions. A set of requirements for Milestone 2 will be posted after Milestone 1 submission. Teams must cover all the requirements that will be posted.
- The full grading scheme for Milestone 2 will be posted with the requirements.
- Your application should not be just a mere copy of other existing applications.
- Your application's UI should be self-contained. Meaning, that all functionalities needed to be fulfilled should be doable through the system's UI.
- The application's UI must be intuitive and easy to use, as it should follow the basic UX/UI guidelines outlined in lectures and tutorials. This means there should be no hurdles while the Product Manager is navigating through the user journeys of your application and it should have a smooth flow.
- If your application crashes and you need to re-run your code, you will lose marks.
- If your application requires you to
- Your application's UI must be coherent and consistent (even with your logo), and it must follow a color palette. For color palette examples you can check <https://www.canva.com/colors/color-palette-generator> among other online resources.
- The grading of your system will **purely subjective based on the opinion and viewpoint of your assigned Product Manager** and will consider the degree of:
 1. Learnability.
 2. Visibility.
 3. Efficiency.
 4. Design for errors.
 5. Provision of Feedback.
 6. Overall satisfaction of the different aspects of your design.
 7. Navigation from page to page and navigation reversibility.

8. Responsiveness of the UI.
9. Consistency of the UI elements.
10. How professional the overall theme of the system looks.
11. The colors, fonts and other elements used in the UI should reflect the Product Manager's preferences.

6.3 Assignment

The grading criteria will be posted later.

7 Evaluations

7.1 Milestone 1

- Milestone 1 will be graded by the assigned Product Managers and a 1 on 1 review will be carried out for it.
- Teams can choose to all be present or only send 1-2 representatives to the review.
- Dates of the review will be announced later.

7.2 Milestone 2

- Milestone 2 will be evaluated by the Product Manager on **ONE** laptop.
- All team members **MUST** be present.
- Any team member that does not attend the evaluation will receive a **ZERO**.
- Product Manager will conduct a feedback session for teams that want early feedback on their UI/UX before the official evaluations.

7.3 Assignment

- The assignment will be evaluated by the Product Manager at the same time as Milestone 2.
- All team members **MUST** be present.
- Any team member that does not attend the evaluation will receive a **ZERO**.